

# **Gebze Technical University**

## **Dep. of Mechanical Engineering**

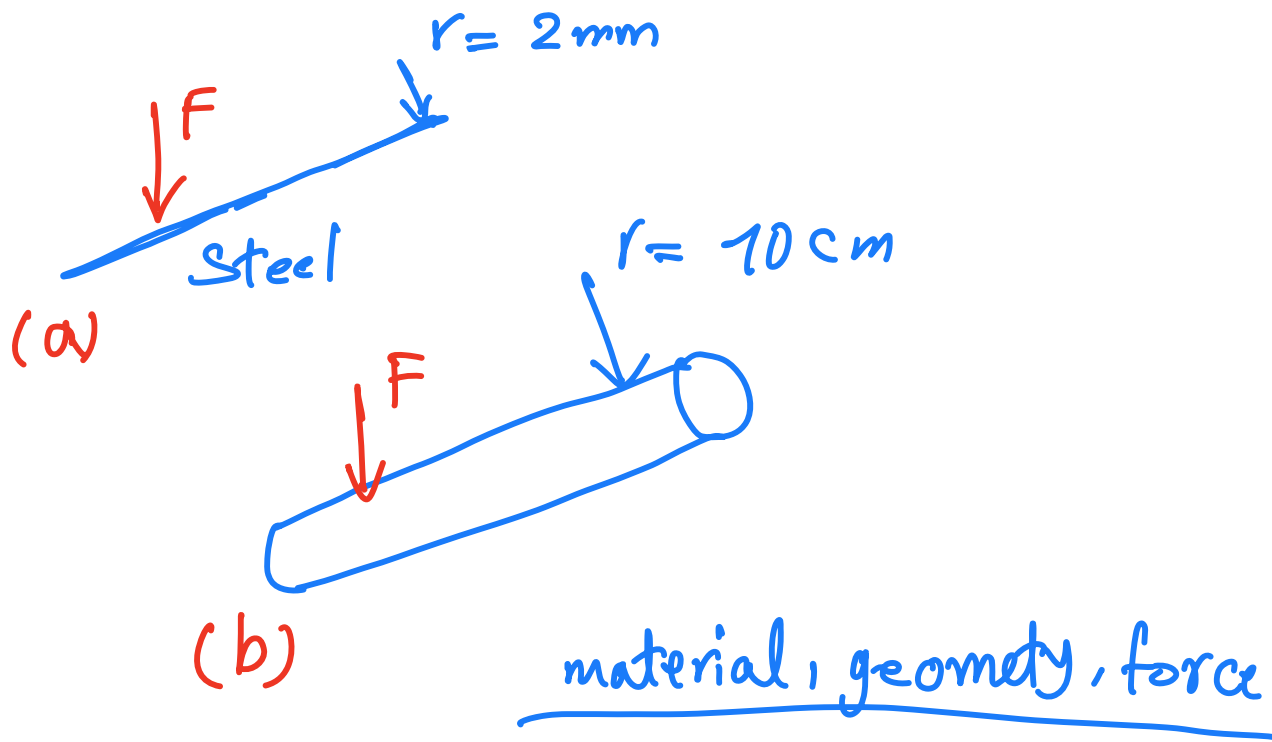
**Statics (ME 231)**

### **Lecture 1: General Principles**

**By Saeed Lotfan**

## Mechanics

- *Rigid-body mechanics*
- *Deformable-body mechanics*
- *Fluid mechanics*



# Mechanics

Why study statics?

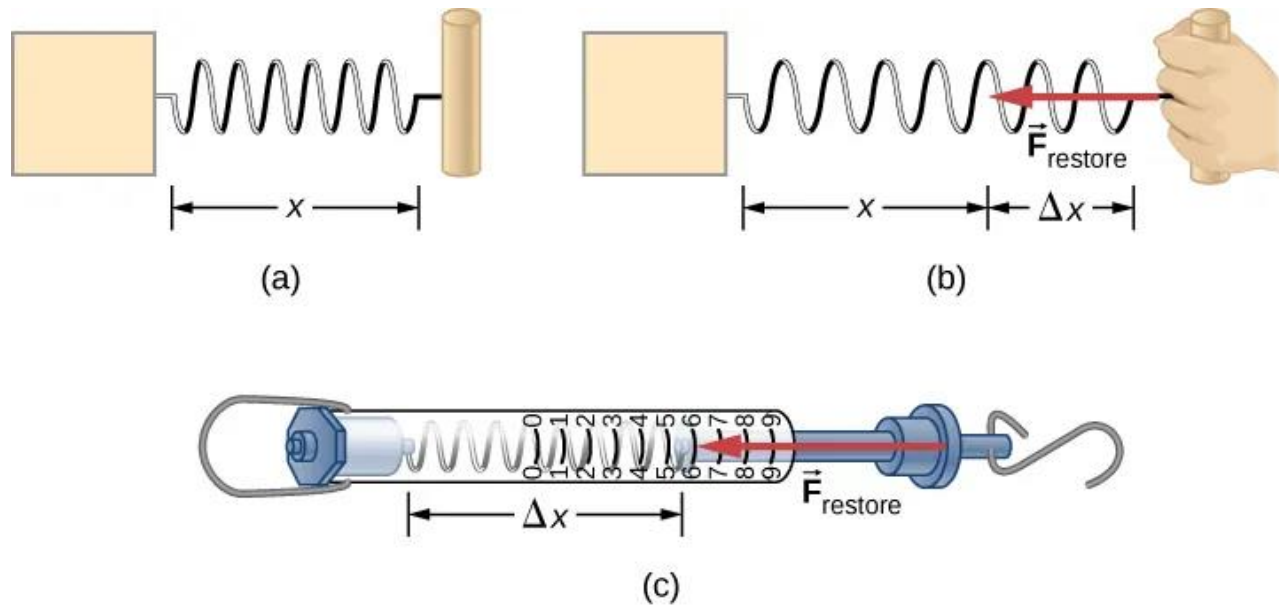
Difference between statics and dynamics?



## Fundamental Concepts

### Basic quantities:

- **Length** (the size of a physical system)
- **Time** (defines the succession of events)
- **Mass** (measure of a quantity of matter)
- **Force** (“push” or “pull” exerted by one body on another, e.g., gravitational, electrical, and magnetic)



## Fundamental Concepts

### Idealization:

- Particle
- Rigid body

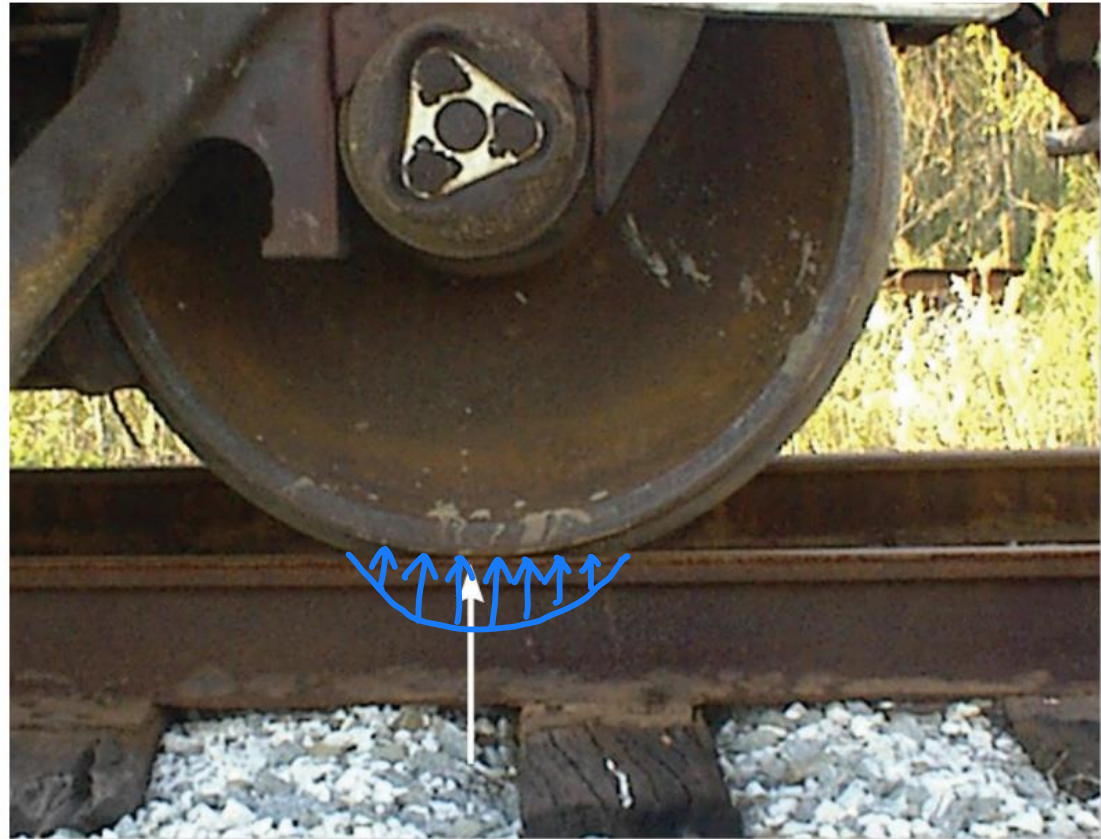


Three forces act on the ring. Since these forces all meet at a point, then for any force analysis, we can assume the ring to be represented as a particle.

## Fundamental Concepts

### Idealization:

- Concentrated force

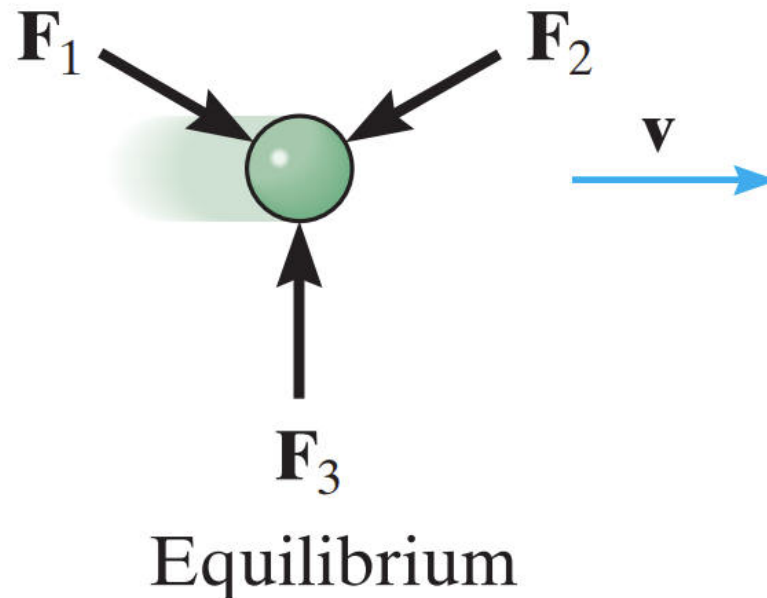


Steel is a common engineering material that does not deform very much under load. Therefore, we can consider this railroad wheel to be a rigid body acted upon by the concentrated force of the rail.

## Fundamental Concepts

### Newton's three laws of motion:

**First Law.** A particle originally at rest, or moving in a straight line with constant velocity, tends to remain in this equilibrium state provided the particle is not subjected to an unbalanced force.



## Fundamental Concepts

### Newton's three laws of motion:

**Second Law.** A particle acted upon by an unbalanced force  $F$  experiences an acceleration  $a$  that has the same direction as the force and a magnitude that is directly proportional to the force. **Or** the unbalanced force acting on the particle is proportional to the time rate of change of the particle's linear momentum.

$$\mathbf{F} = m\mathbf{a}$$

*resistance  
against*

*force/motion*

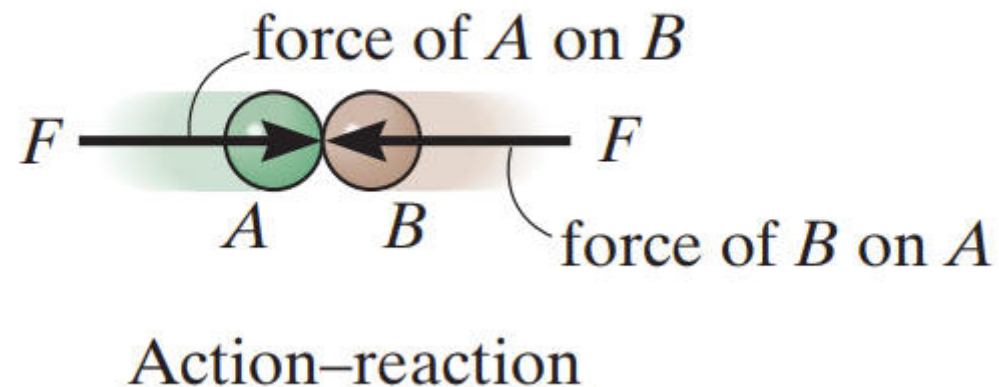


Accelerated motion

## Fundamental Concepts

### Newton's three laws of motion:

**Third Law.** The mutual forces of action and reaction between two particles are equal, opposite, and collinear.



## Fundamental Concepts

### Newton's law of gravitational attraction:

$$F = G \frac{m_1 m_2}{r^2}$$

where

$F$  = force of gravitation between the two particles

$G$  = universal constant of gravitation; according to experimental evidence,  $G = 66.73(10^{-12}) \text{ m}^3/(\text{kg} \cdot \text{s}^2)$

$m_1, m_2$  = mass of each of the two particles

$r$  = distance between the two particles



## Fundamental Concepts

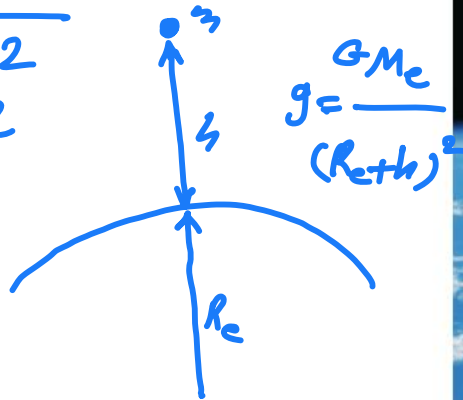
Newton's law of gravitational attraction:

What is weight then?

$$F = W = G \frac{mM_e}{R_e^2}$$

$$W = mg ;$$

$$g = \frac{GM_e}{R_e^2}$$



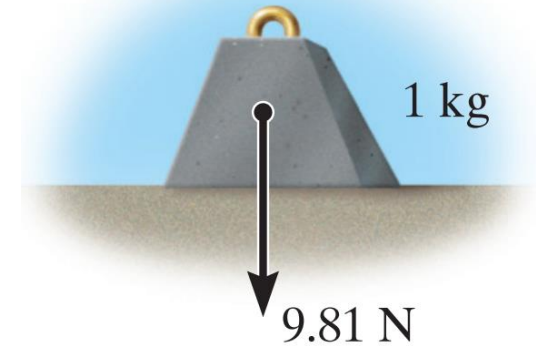
The astronaut's weight is diminished since she is far removed from the gravitational field of the earth.

## International System of Units (SI)

### Base units and compatibility:

$$F = ma \quad \left( \text{kg} \cdot \frac{\text{m}}{\text{s}^2} \right) \equiv \text{N}$$

$$W = F \cdot d \quad \left( \text{kg} \frac{\text{m}}{\text{s}^2} \cdot \text{m} \right) = \left( \text{kg} \frac{\text{m}^2}{\text{s}^2} \right) = \text{J} = \text{N} \cdot \text{m}$$



| TABLE 1-1 International System of Units |        |        |          |                                                                         |
|-----------------------------------------|--------|--------|----------|-------------------------------------------------------------------------|
| Quantity                                | Length | Time   | Mass     | Force                                                                   |
| SI Units                                | meter  | second | kilogram | newton*                                                                 |
|                                         | m      | s      | kg       | $\text{N}$ $\left( \frac{\text{kg} \cdot \text{m}}{\text{s}^2} \right)$ |
| *Derived unit.                          |        |        |          |                                                                         |

## International System of Units (SI)

**Prefixes:** When a numerical quantity is either very large or very small, the SI units used to define its size may be modified by using a prefix.

| TABLE 1–2 Prefixes |                  |        |           |
|--------------------|------------------|--------|-----------|
|                    | Exponential Form | Prefix | SI Symbol |
| <i>Multiple</i>    |                  |        |           |
| 1 000 000 000      | $10^9$           | giga   | G         |
| 1 000 000          | $10^6$           | mega   | M         |
| 1 000              | $10^3$           | kilo   | k         |
| <i>Submultiple</i> |                  |        |           |
| 0.001              | $10^{-3}$        | milli  | m         |
| 0.000 001          | $10^{-6}$        | micro  | $\mu$     |
| 0.000 000 001      | $10^{-9}$        | nano   | n         |

## International System of Units (SI)

### Prefixes:

What is the value of (50 kN)(60 nm)?

$$(50 \times 10^3 \text{ N})(60 \times 10^{-9} \text{ m})$$

$$3000 \times 10^{-6} \text{ N}\cdot\text{m} = 3 \times 10^{-3} \text{ J} = 3 \text{ mJ}$$

## Numerical Calculations

Reporting data with justifiable accuracy using appropriate significant figures:

- **Dimensional Homogeneity**

(e.g.,  $s = vt + 1/2 at^2$ , or  $a = 2s/t^2 - 2v/t$ )

$$[m] \equiv \left[\frac{m}{s}\right] \cdot s + [-] \left[\frac{m}{s^2}\right] s^2$$

$$[m] \equiv [m] + [m]$$

## Numerical Calculations

Reporting data with justifiable accuracy using appropriate significant figures:

- **Significant Figures:** The number of significant figures contained in any number determines the accuracy of the number.

**What is engineering notation for 23400 or 0.00025300 with three or five significant figures?**

$$234 \times 10^2 = 2.34 \times 10^5$$
$$23400 = 2.3400 \times 10^5$$

$$2.53 \times 10^{-4}$$
$$2.5300 \times 10^{-4}$$

## Numerical Calculations

Reporting data with justifiable accuracy using appropriate significant figures:

- **Rounding Off Numbers:** Rounding off a number is necessary so that the accuracy of the result will be the same as that of the problem data.

**Examples for round off the numbers below to three significant figures:**

3.558,    4.412,    8.265,    6.275,    2.255

3.5 6  $\square$   
 $\downarrow$   
 $\times$

4.41

8.26

6.28

2.26

$< 5$

$x \rightarrow x$

$> 5$

$x \rightarrow x+1$

$= 5$

$x \rightarrow \begin{cases} \text{odd} \rightarrow x+1 \\ \text{even} \rightarrow x \end{cases}$

## Numerical Calculations

**Note to calculations:** When a sequence of calculations is performed, it is best to store the intermediate results in the calculator. In other words, do not round off calculations until expressing the final result. This procedure maintains precision throughout the series of steps to the final solution.

## Concluding Notes

- Statics is the study of bodies that are at rest or move at constant velocity.
- A particle has a mass but a size that can be neglected, and a rigid body does not deform under load.
- A force is considered as a “push” or “pull” of one body on another.
- Concentrated forces are assumed to act at a point on a body.
- Newton’s three laws of motion should be memorized.
- Mass is a measure of a quantity of matter that does not change from one location to another. Weight refers to the gravitational attraction of the earth on a body or quantity of mass. Its magnitude depends upon the elevation at which the mass is located.

## Concluding Notes

- In the SI system the unit of force, the newton, is a derived unit. The meter, second, and kilogram are base units.
- Prefixes G, M, k, m,  $\mu$ , and n are used to represent large and small numerical quantities. Their exponential size should be known, along with the rules for using the SI units.
- Perform numerical calculations with several significant figures, and then report the final answer to three significant figures.
- Algebraic manipulations of an equation can be checked in part by verifying that the equation remains dimensionally homogeneous.
- Know the rules for rounding off numbers.

## Practice at Home

1. Convert 100 km/h to m/s and 24 m/s to km/h.
2. Convert the density of steel 7.85 g/cm<sup>3</sup> to kg/m<sup>3</sup>.
3. Evaluate each of the following and express with SI units having an appropriate prefix:
  - (a) (50 mN)(6 GN),
  - (b) (400 mm)(0.6 MN)<sup>2</sup>,
  - (c) 45 MN<sup>3</sup>/900 Gg.

Remember to round off the final answer to three significant figures.